Dwell-Conditioned Cycle-to-Cycle Switching-Time Feedback for Mode-Dependent Switched-Affine Systems

Daniel Vieira[1*] and Roberto Kawakami Harrop Galvão[1]

[1*]Systems and Control, Instituto Tecnológico de Aeronáutica, São José dos Campos, São Paulo, Brazil.

*Corresponding author(s). E-mail(s): danielvie@gmail.com;

Abstract

This paper adresses the problem of controlling a switching-time affine system under dwell-time constraints. The goal is to steer the system to a target periodic trajectory rather than a constant average operating point. We propose a proportional controller and a signal conditioner to comply with the minimum dwell-time constraints. The method combines a solver-free enforcement of the minimum dwell-time constraints with a common quadratic Lyapunov function approach. The controller acts by generating the delta time to be applyed on the switching instants, then conditioning the signal by reducing the requested adjustiments when the schedule is not feasible. The example used here is a three-cell DC-DC converter, we verify the stability of the linearized model using this approach. The simulations show faster convergence of this method compared to a conservative feedback and open-loop opearation while respecting the dwell-time contraints. The analysis guarantees exponential stability of the lineared model even when the conditioning factor changes from cyclo to other.

Keywords: switched-affine systems, periodic trajectories, switching-time control, dwell-time conditioning, common quadratic Lyapunov function, semidefinite programming

1 Introduction

Power converters control voltage and current by moving through a sequence of switch configurations. The switching pattern must respect electrical limits, switching frequency, a minimum dwell duration for transition, and a short computation time Karamanakos et al. (2020). Keeping the individual configurations explicit gives a hybrid-affine or switched-affine model Albea-Sanchez et al. (2021). When repeating the pattern, then it produces a periodic trajectory rather than a constant state, and controlling such trajectories is an established problem for switched-affine systems Egidio et al. (2020); Xu and Lazar (2024).

Predictive control tackles the problem by solving a constrained optimization problem online. Recent work reduces the online burden through Laguerre parameterization, dedicated fixed-frequency solvers, and candidate pruning Saeed et al. (2022); Yang et al. (2022); Augustine and Patil (2023). Controllers also differ in what they may change. State-dependent switching laws choose the active subsystem Egidio et al. (2022), where switching-time control keeps an ordered sequence and moves its transition instants. We take this same approach by changing the transition instants while keeping all other parameters fixed. At each cycle start, the error is computed and the offsets applied during that cycle, with no online solver.

Fixed-period switching-time control shifts only the interior switching instants without changing the mode sequence or cycle period. It targets a reference periodic trajectory rather than an equilibrium point. Patino et al. studied how individual switching instants affect mode-dependent cyclic systems and used predictive control to track an optimal periodic trajectory Patino et al. (2010). Marcolino et al. regulated the cycle-start state toward a target point on the periodic trajectory, referred to in this paper as the cycle anchor. For a fixed-period linear model with switched actuator levels, they used a one-cycle model to compute switching-instant corrections through discrete LQR and dwell-constrained predictive control Marcolino et al. (2021). Shifting a switching instant makes the preceding dwell duration larger and the following one shorter, coupling their minimum-dwell constraints. Marcolino et al. enforce these constraints through online optimization in their predictive controller. Their LQR method instead clips each adjustment separately to meet the dwell limits. This can change the direction of the requested adjustments and may shift the final switching instant Marcolino et al. (2021). Preserving the direction of the control action through control allocation still requires online optimization and does not address fixed-period dwell constraints Nakamura-Zimmerer and Miller (2026). The remaining problem is to compute the largest feasible scale without an online solver. The scale must preserve the action direction and mode order without changing the cycle period.

To make fixed-period switching-time feedback dwell-feasible, we scale every requested offset by the same factor, taking the smallest of 1 and the bounds imposed by the N dwell constraints. We find this largest feasible factor in $O(N)$ time without an online solver, while scaling the full raw offset vector preserves its direction, the mode order, and the cycle period. We assess convergence separately with an offline common quadratic Lyapunov certificate, which establishes uniform exponential stability of the conditioned linearized model even when the factor changes between cycles. The conditioner makes each finite timing request dwell-feasible, but the certificate does not guarantee convergence of the exact nonlinear plant from arbitrary errors. Section 2 derives the cycle model, section 3 develops the conditioner, section 4 presents the stability analysis, and section 5 presents the converter study.

2 Problem formulation and one-cycle model

The control objective is to approach a prescribed periodic trajectory without changing its mode sequence or cycle period. We first express this objective at cycle starts, then derive how shifts of the interior switching instants change the next cycle-start state.

2.1 Plant, reference trajectory, and timing coordinates

Consider the switched-affine plant

$$\dot{\boldsymbol{x}}(t) = A_{\sigma(t)}\boldsymbol{x}(t) + \boldsymbol{b}_{\sigma(t)}, \qquad \boldsymbol{x}(t) \in \mathbb{R}^n. \quad (1)$$

Every cycle visits the fixed sequence $(q_1, \dots, q_N)$. Its nominal cycle-relative boundaries and dwell durations are

$$\begin{aligned} 0 = \bar{t}_0 < \bar{t}_1 < \cdots < \bar{t}_N = T, \\ \bar{d}_i = \bar{t}_i - \bar{t}_{i-1}. \end{aligned} \quad (2)$$

Repeating this schedule generates the reference periodic trajectory $\bar{\boldsymbol{x}}(t)$. Its cycle anchor is $\bar{\boldsymbol{x}}(0) = \bar{\boldsymbol{x}}(T)$. With $\boldsymbol{x}_k = \boldsymbol{x}(kT)$, the cycle-start error is $\boldsymbol{e}_k = \boldsymbol{x}_k - \bar{\boldsymbol{x}}(0)$. The reference schedule and trajectory are given design data. The target is this phase-aligned trajectory, not a constant average operating point.

The controller moves only the $N-1$ interior instants:

$$\begin{aligned} t_{k,i} = \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N-1, \\ \delta\tau_{k,0} = \delta\tau_{k,N} = 0. \end{aligned} \quad (3)$$

During interval i of cycle k, the plant uses mode q_i between the absolute times $kT + t_{k,i-1}$ and $kT + t_{k,i}$. The applied dwell vector satisfies

$$\begin{aligned} \boldsymbol{d}_k = \bar{\boldsymbol{d}} + \delta\boldsymbol{d}_k, \qquad \delta\boldsymbol{d}_k = D\delta\boldsymbol{\tau}_k, \\ D_{ij} = \begin{cases} 1, & i = j, \\ -1, & i = j+1, \\ 0, & \text{otherwise}, \end{cases} \end{aligned} \quad (4)$$

where $D \in \mathbb{R}^{N\times(N-1)}$. Thus delaying one instant lengthens the preceding interval and shortens the following one by the same amount. Since $\mathbf{1}^\mathsf{T} D = 0$, every offset vector preserves T. The remaining requirement is $\boldsymbol{d}_k \geq d_{\min}\mathbf{1}$ for a specified $d_{\min} > 0$.

Figure 1(a)–(b) illustrates the timing coordinates using the converter schedule introduced in section 5. In panel (b), delaying instant 8 by 12 µs lengthens dwell 8 and shortens dwell 9 by the same amount, with all other offsets zero. The shortened dwell remains above the assumed 3 µs bound. This isolated offset illustrates the dwell coupling; panel (c) instead shows the aggressive controller's first-cycle raw request.

2.2 Exact augmented propagation and terminal error

Introduce the augmented state and matrix

$$\mathcal{X}(t) = \begin{bmatrix} \boldsymbol{x}(t) \\ 1 \end{bmatrix}, \qquad \widetilde{A}_q = \begin{bmatrix} A_q & \boldsymbol{b}_q \\ \mathbf{0}^\mathsf{T} & 0 \end{bmatrix}. \tag{5}$$

The affine system becomes the homogeneous system

$$\dot{\mathcal{X}}(t) = \widetilde{A}_{\sigma(t)}\mathcal{X}(t). \tag{6}$$

For this derivation, consider one cycle in cycle-relative time. Suppress the cycle index by writing $t_i = t_{k,i}$ and $\boldsymbol{x}(t)$ for $\boldsymbol{x}(kT + t)$. The endpoints remain fixed: $t_0 = \bar{t}_0 = 0$ and $t_N = \bar{t}_N = T$. The mode on interval $[t_{i-1}, t_i)$ is q_i.

For each interval, define

$$\begin{aligned} \Delta t_i &= t_i - t_{i-1}, \\ \Delta \bar{t}_i &= \bar{t}_i - \bar{t}_{i-1}, \\ \Delta t_i &= \Delta \bar{t}_i + \delta t_i, \qquad F_i = \widetilde{A}_{q_i}. \end{aligned} \tag{7}$$

Here $\Delta t_i = d_{k,i}$, $\Delta \bar{t}_i = \bar{d}_i$, and $\delta t_i = \delta d_{k,i}$. Thus δt_i is a duration change, not a switching-instant offset. The state transition on interval i is

$$\mathcal{X}(t_i) = e^{F_i \Delta t_i}\mathcal{X}(t_{i-1}). \tag{8}$$

Therefore, the exact cycle propagation is

$$\mathcal{X}(t_N) = e^{F_N \Delta t_N} e^{F_{N-1}\Delta t_{N-1}} \cdots e^{F_1 \Delta t_1}\mathcal{X}(t_0). \tag{9}$$

The first interval is the rightmost factor. Matrices belonging to different modes cannot generally be interchanged.

Define the nominal transition matrix for interval i by

$$\phi_i = e^{F_i \Delta \bar{t}_i}. \tag{10}$$

The nominal augmented trajectory is $\bar{\mathcal{X}}(t) = [\bar{\boldsymbol{x}}(t)^\mathsf{T}\ 1]^\mathsf{T}$, so its terminal state satisfies

$$\bar{\mathcal{X}}(t_N) = \phi_N \phi_{N-1} \cdots \phi_1 \bar{\mathcal{X}}(t_0). \tag{11}$$

Both terminal states are evaluated at the same fixed time T.

Let

$$\begin{aligned} \boldsymbol{e}(t) &= \mathcal{X}(t) - \bar{\mathcal{X}}(t), \\ \mathcal{X}(t_0) &= \bar{\mathcal{X}}(t_0) + \boldsymbol{e}(t_0). \end{aligned} \tag{12}$$

Here $\boldsymbol{e}(t) \in \mathbb{R}^{n+1}$ is the augmented error, whose last component is zero. The physical cycle-start error $\boldsymbol{e}_k \in \mathbb{R}^n$ contains only the first n components at the cycle start. Subtracting (11) from (9) gives the exact terminal error

$$\begin{aligned} \boldsymbol{e}(t_N) = {} & e^{F_N(\Delta \bar{t}_N + \delta t_N)} e^{F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \cdots \\ & e^{F_1(\Delta \bar{t}_1 + \delta t_1)}\big(\bar{\mathcal{X}}(t_0) + \boldsymbol{e}(t_0)\big) \\ & - \phi_N \phi_{N-1} \cdots \phi_1 \bar{\mathcal{X}}(t_0). \end{aligned} \tag{13}$$

2.3 Linearization in the interval durations

For a constant matrix F_i, expand $e^{F_i s}$ around $s = \Delta \bar{t}_i$:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}s} e^{F_i s} &= F_i e^{F_i s}, \\ e^{F_i(\Delta \bar{t}_i + \delta t_i)} &= e^{F_i \Delta \bar{t}_i} + F_i e^{F_i \Delta \bar{t}_i}\,\delta t_i + \mathcal{O}(\delta t_i^2) \\ &= \phi_i + F_i \phi_i\,\delta t_i + \mathcal{O}(\delta t_i^2) \\ &\doteq \phi_i + F_i \phi_i\,\delta t_i. \end{aligned} \tag{14}$$

Here $\doteq$ denotes equality up to first order around zero initial error and zero duration changes. Terms containing two duration changes or a duration change multiplied by the initial error are second order.

Substituting the first-order approximation into every transition gives

$$\begin{aligned} & e^{F_N(\Delta \bar{t}_N + \delta t_N)} e^{F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \cdots \\ & \quad e^{F_1(\Delta \bar{t}_1 + \delta t_1)} \\ & \doteq (\phi_N + F_N \phi_N\,\delta t_N)(\phi_{N-1} + F_{N-1}\phi_{N-1}\,\delta t_{N-1}) \\ & \qquad \cdots (\phi_1 + F_1 \phi_1\,\delta t_1). \end{aligned} \tag{15}$$

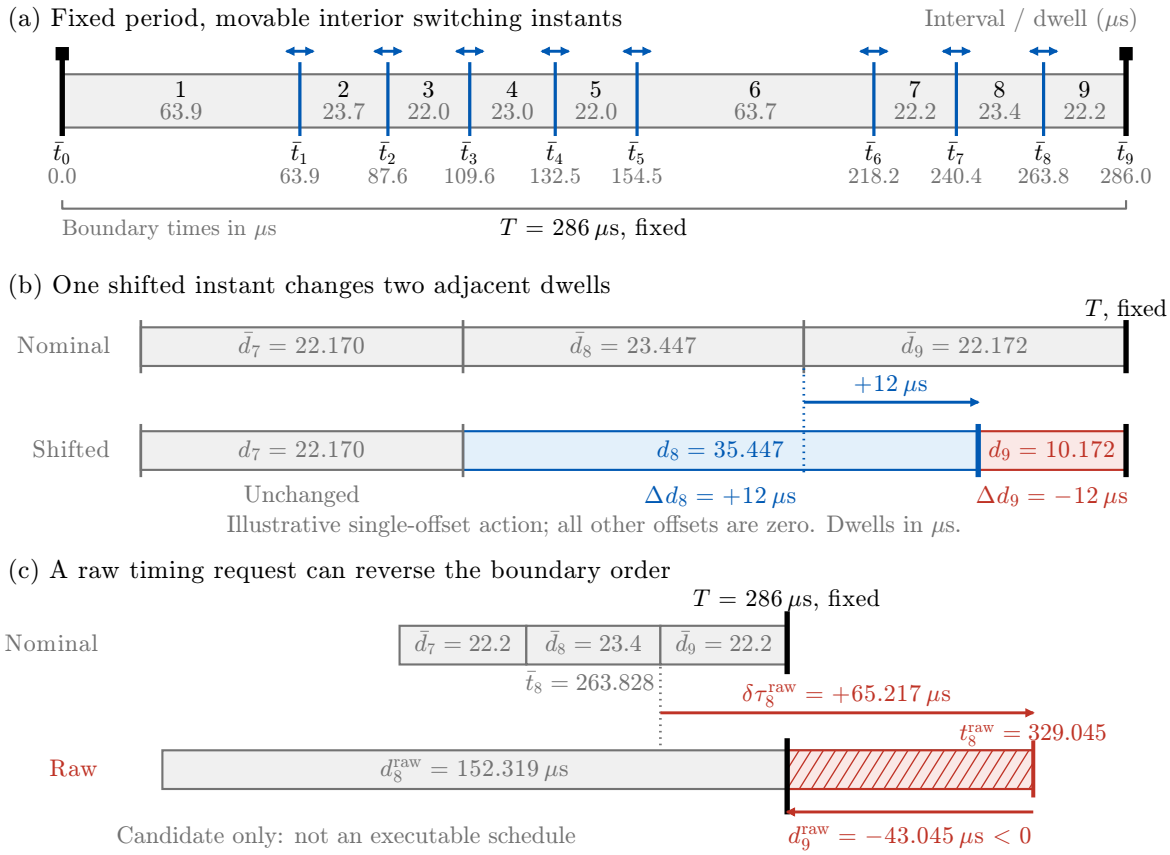


Figure 1 Switching-instant adjustments in the nine-interval schedule. (a) Nominal schedule with eight adjustable interior switching instants. (b) Illustrative +12 µs offset of instant 8. (c) Inadmissible first-cycle raw request from the aggressive controller. Hatching denotes a negative candidate dwell.

Multiplying these factors while keeping at most one duration change per term yields

$$\begin{aligned}(\phi_N + F_N\phi_N\,\delta t_N)\cdots(\phi_1 + F_1\phi_1\,\delta t_1)\\ \doteq \phi_N\phi_{N-1}\cdots\phi_1\\ + F_N\phi_N\phi_{N-1}\cdots\phi_1\,\delta t_N\\ + \phi_N F_{N-1}\phi_{N-1}\cdots\phi_1\,\delta t_{N-1}\\ + \cdots\\ + \phi_N\phi_{N-1}\cdots\phi_2 F_1\phi_1\,\delta t_1.\end{aligned} \tag{16}$$

Each inserted F_i stays next to its own nominal transition ϕ_i, with the other interval transitions in their original order.

Substitute (16) into (13). The nominal term $\phi_N\cdots\phi_1\bar{\mathcal{X}}(t_0)$ cancels. Discarding the products $\delta t_i \boldsymbol{e}(t_0)$ then gives

$$\begin{aligned}\boldsymbol{e}(t_N) \doteq{}& \phi_N\phi_{N-1}\cdots\phi_1\boldsymbol{e}(t_0)\\ &+ \phi_N\cdots\phi_2F_1\phi_1\bar{\mathcal{X}}(t_0)\,\delta t_1\\ &+ \phi_N\cdots\phi_3F_2\phi_2\phi_1\bar{\mathcal{X}}(t_0)\,\delta t_2\\ &+\cdots\\ &+ F_N\phi_N\phi_{N-1}\cdots\phi_1\bar{\mathcal{X}}(t_0)\,\delta t_N.\end{aligned} \tag{17}$$

Collect the duration changes and their sensitivity columns:

$$\delta\boldsymbol{t} = \begin{bmatrix}\delta t_1 & \delta t_2 & \cdots & \delta t_N\end{bmatrix}^{\mathsf{T}}, \tag{18}$$

$$\begin{aligned}\Gamma = \Big[&\phi_N\cdots\phi_2F_1\phi_1\bar{\mathcal{X}}(t_0),\\ &\phi_N\cdots\phi_3F_2\phi_2\phi_1\bar{\mathcal{X}}(t_0),\ \ldots,\\ &F_N\phi_N\phi_{N-1}\cdots\phi_1\bar{\mathcal{X}}(t_0)\Big].\end{aligned} \tag{19}$$

The columns are concatenated horizontally in the same order as the entries of $\delta \boldsymbol{t}$, giving $\Gamma \in \mathbb{R}^{(n+1)\times N}$. For the final compact form, define the complete nominal transition $\Phi \in \mathbb{R}^{(n+1)\times(n+1)}$ by

$$\Phi = \phi_N \phi_{N-1} \cdots \phi_1. \tag{20}$$

Equation (17) is thus

$$\boldsymbol{e}(t_N) \doteq \Phi \boldsymbol{e}(t_0) + \Gamma \delta \boldsymbol{t}. \tag{21}$$

2.4 Physical model and independent switching instants

We now extract the physical error and return to cycle-indexed notation. Subtracting two augmented states gives an error with last coordinate zero. Define

$$\Pi = \begin{bmatrix} I_n & \mathbf{0} \end{bmatrix}, \qquad \iota = \begin{bmatrix} I_n \\ \mathbf{0}^\mathsf{T} \end{bmatrix}. \tag{22}$$

Periodicity gives $\bar{\boldsymbol{x}}(t_N) = \bar{\boldsymbol{x}}(t_0) = \bar{\boldsymbol{x}}(0)$, so $\boldsymbol{e}(t_0) = \iota \boldsymbol{e}_k$ and $\Pi \boldsymbol{e}(t_N) = \boldsymbol{e}_{k+1}$. With $\delta \boldsymbol{t} = \delta \boldsymbol{d}_k$, projection of (21) gives

$$\begin{aligned} \boldsymbol{e}_{k+1} &\doteq \Phi_x \boldsymbol{e}_k + G_d \delta \boldsymbol{d}_k, \\ \Phi_x &= \Pi \Phi \iota = e^{A_{q_N} \bar{d}_N} \cdots e^{A_{q_1} \bar{d}_1}, \\ G_d &= \Pi \Gamma. \end{aligned} \tag{23}$$

The physical nominal cycle matrix is $\Phi_x \in \mathbb{R}^{n\times n}$, whereas Φ remains the augmented transition. The physical dwell-sensitivity matrix is $G_d \in \mathbb{R}^{n\times N}$. From here onward, $\boldsymbol{e}_k$, or $\boldsymbol{e}$ when the cycle index is omitted, denotes the physical error in $\mathbb{R}^n$.

The N duration changes are not independent under the fixed-period constraint. They sum to zero and satisfy $\delta \boldsymbol{d}_k = D \delta \boldsymbol{\tau}_k$, as defined in (4). Substitution gives the model in the controller's $N-1$ independent switching-instant offsets, with its remainder written explicitly:

$$\begin{aligned} \boldsymbol{e}_{k+1} &= \Phi_x \boldsymbol{e}_k + \Gamma_\tau \delta \boldsymbol{\tau}_k \\ &\quad + \mathcal{O}\big((\|\boldsymbol{e}_k\| + \|\delta \boldsymbol{\tau}_k\|)^2\big), \\ \Gamma_\tau &= G_d D. \end{aligned} \tag{24}$$

Here $\Gamma_\tau \in \mathbb{R}^{n\times(N-1)}$ maps the independent timing offsets to the physical error. The remainder is understood in fixed state and timing units. The exact physical one-cycle map underlying this approximation is

$$\begin{aligned} \mathcal{P}(\boldsymbol{x}, \delta \boldsymbol{\tau}) = \Pi e^{F_N[\bar{d}_N + (D\delta\boldsymbol{\tau})_N]} \cdots \\ e^{F_1[\bar{d}_1 + (D\delta\boldsymbol{\tau})_1]} \begin{bmatrix} \boldsymbol{x} \\ 1 \end{bmatrix}. \end{aligned} \tag{25}$$

The nominal schedule satisfies $\mathcal{P}(\bar{\boldsymbol{x}}(0), \mathbf{0}) = \bar{\boldsymbol{x}}(0)$. The augmented nominal product has the block form

$$\Phi = \begin{bmatrix} \Phi_x & \boldsymbol{h} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}. \tag{26}$$

Its unit eigenvalue comes from the auxiliary constant coordinate. Stability is tested on the physical error dynamics, not on the full augmented matrix.

Each column of D subtracts two adjacent dwell-sensitivity columns. Factoring this difference gives

$$\begin{aligned} \Gamma_{\tau,i} = e^{A_{q_N} \bar{d}_N} \cdots e^{A_{q_{i+1}} \bar{d}_{i+1}} \\ \times \big[(A_{q_i} - A_{q_{i+1}}) \bar{\boldsymbol{x}}(\bar{t}_i) + \boldsymbol{b}_{q_i} - \boldsymbol{b}_{q_{i+1}}\big] \end{aligned} \tag{27}$$

The bracket is the change in the affine vector field caused by trading time between adjacent modes. The remaining transitions carry this effect to the cycle end. This recovers the switching-sensitivity structure in Patino et al. (2010); switching-time sensitivities also appear in the hybrid limit-cycle analysis of Flieller et al. (2006).

For the special case $A_{q_i} = A_c$ and $\boldsymbol{b}_{q_i} = B_c \boldsymbol{u}_{i-1}$,

$$\Phi_x = e^{A_c T}, \quad \Gamma_{\tau,i} = e^{A_c(T-\bar{t}_i)} B_c (\boldsymbol{u}_{i-1} - \boldsymbol{u}_i), \tag{28}$$

which is the fixed-period switched-actuator model of Marcolino et al. (2021), with the corresponding input-jump sign convention. Mode-dependent state matrices require the full expression in (27).

2.5 Raw proportional timing feedback

A fixed gain $K \in \mathbb{R}^{(N-1)\times n}$ requests

$$\delta \boldsymbol{\tau}_k^{\text{raw}} = -K \boldsymbol{e}_k, \qquad A_{\text{cl}} = \Phi_x - \Gamma_\tau K. \tag{29}$$

The gain can be designed from (24), for example by discrete LQR. Here the gain is fixed before conditioning and certification. Although a Schur A_{cl} gives a locally stabilizing raw law, it does not make every requested schedule admissible. In figure 1(c),

the raw request moves the last interior instant beyond T, making $d_N = T - t_{k,N-1} < 0$. A minimum dwell constraint is stricter still: it requires $t_{k,N-1} \le T - d_{\min}$ even before the boundaries cross. The next section enforces all such adjacent-dwell constraints; section 4 examines the resulting conditioned dynamics.

3 Solver-free dwell-time conditioning

The conditioner determines how much of the raw timing action can be applied without making any dwell duration shorter than $d_{\min}$. It uses one factor $\beta_k \in [0, 1]$ for all switching-instant offsets. A factor of one keeps the complete request; a smaller factor moves each requested instant back toward its nominal position. Because moving one instant changes two adjacent dwell durations, the factor must respect all intervals.

3.1 Computing the conditioning factor

First, calculate how much each nominal interval can be shortened. This available margin is

$$\boldsymbol{m} = \bar{\boldsymbol{d}} - d_{\min}\mathbf{1} > \mathbf{0}. \tag{30}$$

We assume strictly positive margins, so the nominal schedule satisfies every dwell constraint before control is applied.

Next, take adjacent differences of the raw switching-instant offsets to obtain the requested dwell changes,

$$\boldsymbol{q}_k = D\delta\boldsymbol{\tau}_k^{\mathrm{raw}}.$$

If $q_{k,i} \ge 0$, interval i stays unchanged or becomes longer and imposes no restriction. If $q_{k,i} < 0$, the request shortens that interval by $-q_{k,i}$. After scaling, the shortening is $\beta(-q_{k,i})$ and must not exceed m_i. Thus this interval allows at most the fraction $m_i/(-q_{k,i})$ of the raw action. For example, a requested shortening of twice the available margin limits the factor to one half.

The largest factor allowed by all intervals is

$$\begin{aligned} \beta_k &= \min\left\{1,\ \min_{i:q_{k,i}<0} \frac{m_i}{-q_{k,i}}\right\}, \\ \delta\boldsymbol{\tau}_k^{\mathrm{app}} &= \beta_k \delta\boldsymbol{\tau}_k^{\mathrm{raw}}. \end{aligned} \tag{31}$$

The following scan evaluates this factor without trial schedules or an optimization solver:

$\beta_k \leftarrow 1$
for $i = 1, \ldots, N$ **do**
  if $q_{k,i} < 0$ **then**
    $limit \leftarrow m_i/(-q_{k,i})$
    $\beta_k \leftarrow \min\{\beta_k, limit\}$
  end if
end for
$\delta\boldsymbol{\tau}_k^{\mathrm{app}} \leftarrow \beta_k \delta\boldsymbol{\tau}_k^{\mathrm{raw}}$

A feasible raw request leaves $\beta_k = 1$. Otherwise, the smallest ratio reduces the complete action until at least one dwell reaches $d_{\min}$. The empty minimum in (31) is taken as $+\infty$, consistent with leaving the factor at one when no interval is shortened.

3.2 Feasibility and maximality

Proposition 1 (Feasibility and maximality)**.** *Under (30), the factor in (31) satisfies*

$$\bar{\boldsymbol{d}} + \beta_k D\delta\boldsymbol{\tau}_k^{\mathrm{raw}} \ge d_{\min}\mathbf{1}.$$

It is the largest factor in $[0, 1]$ that satisfies all dwell constraints. For a finite raw action, $\beta_k > 0$. The conditioner preserves the direction of a nonzero raw action, the mode order, and the cycle period. The factor can be computed in $O(N)$ time without an online solver.

Proof Each shortened interval imposes the upper bound derived above. Their minimum, capped at one, satisfies every bound. Any larger factor in $[0, 1]$ would violate at least one of them. This proves feasibility and maximality.

Every ratio $m_i/(-q_{k,i})$ used in (31) is positive for a finite raw action. Hence $\beta_k > 0$, and multiplying the complete vector by this factor preserves its direction. The identity $\mathbf{1}^{\mathsf{T}} D = 0$ ensures that the dwell changes sum to zero, so the period remains T. Every applied dwell is at least $d_{\min} > 0$, so the cycle boundaries remain ordered.

Finally, $\boldsymbol{q}_k$ is computed by adjacent differences of the raw offsets. A single scan through the N dwell changes finds the smallest bound. Both operations take $O(N)$ time. □

3.3 Online execution and admissible region

In the cycle model used here, the controller reads the cycle-start state, computes the error and raw offsets, then runs the scan above. It calculates one β_k for the complete schedule, not a new factor at each switching instant. Only the conditioned schedule is applied to the plant. The model assigns no computation delay to these steps.

The errors for which no reduction is needed form the Raw-Action Admissible Region. Substituting $\delta\boldsymbol{\tau}_k^{\mathrm{raw}} = -K\boldsymbol{e}_k$ into the dwell constraints gives

$$\mathcal{C} = \{\boldsymbol{e} : DK\boldsymbol{e} \le \boldsymbol{m}\}. \tag{32}$$

Inside $\mathcal{C}$, the raw action is feasible and $\beta_k = 1$. Outside it, at least one requested dwell is too short and $\beta_k < 1$. The scan already handles both cases, so no separate region-membership test is needed. Section A compares the feasible regions for smaller fixed factors and checks their invariance under the corresponding linearized dynamics.

4 Lyapunov stability and convex optimization

The conditioner keeps the dwell durations feasible, but feasibility alone does not imply convergence. We now examine the linearized dynamics as the conditioning factor changes. The argument first identifies the two endpoint matrices, then uses one Lyapunov function to cover every factor between them. We then formulate an offline certificate computation for a fixed gain.

4.1 Conditioned matrix family

We work directly with the physical cycle-start error $\boldsymbol{e}_k$. The two endpoint matrices are the nominal cycle matrix and the raw closed-loop matrix:

$$A_0 = \Phi_x, \qquad A_1 = A_{\mathrm{cl}}. \tag{33}$$

The conditioner applies a fraction β_k of the raw action. Substituting this action into the first-order model gives

$$\begin{aligned} \boldsymbol{e}_{k+1} &= A(\beta_k)\boldsymbol{e}_k, \\ A(\beta) &= (1-\beta)A_0 + \beta A_1 = \Phi_x - \beta\Gamma_\tau K. \end{aligned} \tag{34}$$

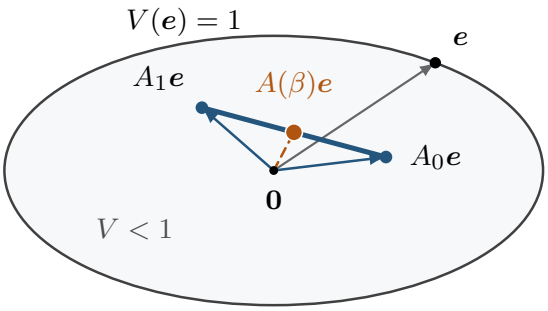


Figure 2 Common-P schematic in error coordinates. Both endpoint images lie inside the unit level set of V. Every intermediate image $A(\beta)\boldsymbol{e}$ lies on the segment joining them.

At $\beta = 0$, the matrix is Φ_x, corresponding to no timing feedback. At $\beta = 1$, it is A_{cl}, corresponding to the complete raw action. Intermediate factors give matrices on the segment joining these endpoints.

Equation (34) is a linearized model. It omits the remainder in (24). We certify it on the closed interval $[0, 1]$. The implemented conditioner returns a positive factor for every finite error, but including zero also covers the limit of arbitrarily weak feedback.

Even if both endpoints are Schur, the matrices between them need not all be stable. Stability of each matrix would also not, by itself, guarantee stability when the factor changes from cycle to cycle. We therefore seek one quadratic Lyapunov function that decreases for both endpoints and remains valid as β_k changes.

4.2 Common quadratic Lyapunov certificate

Using one quadratic function for a family of discrete-time systems is the common quadratic Lyapunov function approach Mason and Shorten (2004). Hu et al. use a common quadratic function to certify contractive ellipsoids for discrete-time actuator saturation by checking decrease at the vertices of a convex family Hu et al. (2002). Here, scaling the complete switching-time action produces a two-endpoint family of linearized one-cycle matrices.

Consider $V(\boldsymbol{e}) = \boldsymbol{e}^{\mathsf{T}}P\boldsymbol{e}$ with $P \succ 0$. The endpoint inequalities below require this function to decrease by at least $\eta\|\boldsymbol{e}\|_2^2$ in one cycle. The same P and the same positive margin η must work at both endpoints.

Theorem 1 (Uniform stability under conditioning)**.** *Suppose there exist* $P = P^{\mathsf{T}} \succ 0$ *and* $\eta > 0$ *such that*

$$P - A_i^{\mathsf{T}} P A_i \succeq \eta I_n, \qquad i \in \{0, 1\}. \tag{35}$$

Then $V(\boldsymbol{e}) = \boldsymbol{e}^{\mathsf{T}} P \boldsymbol{e}$ *decreases for every nonzero state of* (34) *and every* $\beta \in [0, 1]$*. The conditioned linearized model is uniformly exponentially stable for every sequence* $\{\beta_k\} \subset [0, 1]$*, including the state-dependent factors in* (31)*.*

Proof First, express the desired decrease $A(\beta)^{\mathsf{T}} P A(\beta) - P \prec 0$ as a block-matrix inequality. For the same $P \succ 0$ at both endpoints, define

$$\mathcal{S}(\beta) = \begin{bmatrix} -P & A(\beta)^{\mathsf{T}} \\ A(\beta) & -P^{-1} \end{bmatrix}. \tag{36}$$

The lower-right block is negative definite. Its Schur complement is $-P + A(\beta)^{\mathsf{T}} P A(\beta)$, so

$$\mathcal{S}(\beta) \prec 0 \iff A(\beta)^{\mathsf{T}} P A(\beta) - P \prec 0.$$

The endpoint assumptions therefore make both $\mathcal{S}(0)$ and $\mathcal{S}(1)$ negative definite.

Next, use $A(\beta) = (1 - \beta) A_0 + \beta A_1$. Since the same P is used throughout, the diagonal blocks do not change with β. Hence

$$\begin{aligned} \mathcal{S}(\beta) = (1 - \beta) \begin{bmatrix} -P & A_0^{\mathsf{T}} \\ A_0 & -P^{-1} \end{bmatrix} \\ + \beta \begin{bmatrix} -P & A_1^{\mathsf{T}} \\ A_1 & -P^{-1} \end{bmatrix} \prec 0. \end{aligned} \tag{37}$$

Both endpoint blocks are negative definite, and their weights are nonnegative and sum to one. Their convex combination is therefore negative definite for every $\beta \in [0, 1]$.

Taking the Schur complement shows that $V(\boldsymbol{e}) = \boldsymbol{e}^{\mathsf{T}} P \boldsymbol{e}$ decreases strictly for every nonzero error and every $\beta \in [0, 1]$. The same P applies throughout the interval. Since the matrix family depends continuously on β over this compact interval, the decrease has a uniform positive margin. The quadratic Lyapunov criterion therefore establishes uniform exponential stability for any conditioning-factor sequence. □

Figure 2 gives a geometric interpretation. Start with an error on the level set $V(\boldsymbol{e}) = 1$. Both endpoint images lie strictly inside the same ellipsoid. The intermediate image $A(\beta)\boldsymbol{e}$ lies on the line segment joining them, which also stays inside the ellipsoid. Using the same V at every cycle allows this decrease to be carried forward even when β_k changes. The drawing is a two-dimensional schematic, not a converter trajectory or a nonlinear region of attraction.

The proof covers the entire interval, not just sampled factors. It applies to all initial errors of the linearized model, including errors outside $\mathcal{C}$. This does not make the linear approximation valid for arbitrarily large physical errors.

4.3 Offline semidefinite program

Related common-matrix vertex-LMI formulations appear in discrete-time robust stability de Oliveira et al. (1999) and in robust polytopic PWM converter control based on continuous-time averaged models Olalla et al. (2009).

The Lyapunov matrix can be computed offline, after the feedback gain K has been chosen. With $A_0 = \Phi_x$ and $A_1 = A_{\mathrm{cl}}$ fixed, an endpoint SDP is

$$\begin{aligned} \underset{P = P^{\mathsf{T}},\, \eta}{\text{maximize}} \quad & \eta \\ \text{subject to} \quad & \operatorname{tr}(P) = n, \\ & P \succeq \varepsilon_P I_n, \\ & P - A_0^{\mathsf{T}} P A_0 \succeq \eta I_n, \\ & P - A_1^{\mathsf{T}} P A_1 \succeq \eta I_n, \end{aligned} \tag{38}$$

with $\varepsilon_P = 10^{-7}$. The constraint $P \succeq \varepsilon_P I_n$ imposes a positive-definiteness floor. The trace constraint fixes the otherwise arbitrary positive scale of P.

Because the endpoint matrices are fixed, the constraints in (38) are affine in the unknowns P and η. The block form in (36) is affine in β for a fixed P, which is what the proof requires. It is not affine in P because it contains P^{-1}. The numerical application uses the inverse-free inequalities in (38), expressed in scaled coordinates as described in section 5. This optimization finds a certificate for the chosen gain; it does not redesign K.

Successful solver termination is not enough to establish a certificate. We first symmetrize the returned P, then recompute the endpoint decrease matrices and their smallest eigenvalues:

$$\begin{aligned} Q_i &= P - A_i^{\mathsf{T}} P A_i, \qquad i \in \{0, 1\}, \\ \mu &= \min\{\lambda_{\min}(Q_0), \lambda_{\min}(Q_1)\}. \end{aligned} \tag{39}$$

We check that P is positive definite, that its trace has the required value, and that $\mu > 0$. A positive verified margin supplies the common decrease required by theorem 1.

This test is sufficient, not necessary. Including $\beta = 0$ requires the open-loop endpoint A_0 to be Schur. If that endpoint is unstable, the full-interval certificate cannot hold, even if restricting the applied factors might allow stabilization. More generally, failure to find a positive common margin does not establish instability.

The stability theorem applies to the conditioned linearized model. The large-error response of the exact nonlinear system, including the initially conditioned cycles, is evaluated by simulation.

5 Application to a three-cell converter

We now apply the controller to a three-cell converter. After defining the benchmark and gains, we check the cycle model and the numerical Lyapunov certificate. We then compare exact simulations with and without feedback. The benchmark is already open-loop stable, so the comparison asks how much feedback accelerates convergence while keeping every dwell duration feasible.

5.1 Benchmark and controller parameters

We use the three-cell multilevel DC–DC converter benchmark of Patino et al. Patino et al. (2010). The state $\boldsymbol{x} = [v_{C_1}, v_{C_2}, i_L]^\mathsf{T}$ contains the two capacitor voltages and the inductor current. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$, the plant matrices are

$$A_{\boldsymbol{u}} = \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix}, \quad \boldsymbol{b}_{\boldsymbol{u}} = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \tag{40}$$

Both $A_{\boldsymbol{u}}$ and $\boldsymbol{b}_{\boldsymbol{u}}$ depend on the switch state. The common-state-matrix reduction in (28) therefore does not apply. We retain the fixed nine-interval sequence $(0, 1, 3, 7, 2, 0, 4, 7, 4)$. Each physical mode ID is encoded as $4u_1 + 2u_2 + u_3$.

The published cycle boundaries are $(0, 66, 88, 110, 132, 154, 220, 242, 264, 286)$ µs, and the reported anchor is $\bar{\boldsymbol{x}}(0) = (9.9247, 19.2928, 0.9823)^\mathsf{T}$. Propagation with these rounded data does not return exactly to the anchor after one cycle.

Table 1 Converter and timing-feedback parameters. The applied dwell bound is a simulation assumption, not a hardware-calibrated limit.

Quantity	Value
Source E	30 V
Capacitances C_1, C_2	40 µF
Inductance L; resistance R	10 mH; 10 Ω
Intervals N; period T	9; 286 µs
Nominal-design dwell bound	22 µs
Applied dwell bound $d_{\min}$	3 µs
State scale S_x	diag(10, 20, 1)
Timing scale t_s	10 µs
LQR state weight Q_n	I_3
Aggressive input weight R_n	$0.001I_8$
Conservative input weight R_n	$0.1I_8$

We therefore make a one-time least-squares adjustment to the nominal dwell durations. This reconciliation preserves the anchor, the 286 µs period, and the 22 µs nominal-design bound. The resulting boundaries are

$$\bar{\boldsymbol{t}} = (0, 63.889, 87.557, 109.557, 132.538, 154.538, 218.210, 240.380, 263.828, 286)\ \text{µs}. \tag{41}$$

Computations use the full-precision schedule rather than the rounded values displayed here. Its closure error is 1.421e-14 in the infinity norm, and the largest boundary correction is 2.111 µs. This adjustment only prepares the published benchmark for the comparisons. It is not a proposed method for designing reference trajectories.

Figure 1(a) shows the reconciled schedule. Its reference periodic trajectory appears in figures 3 and 5, with the cycle anchor marked in figure 3. The cycle-average states are 10.410 56 V, 19.422 28 V, and 0.956 13 A. Feedback targets the phase-aligned periodic trajectory, not these averages.

5.2 Numerical scaling and feedback gains

For numerical design and reporting, we express the voltage and current errors relative to chosen scales. Let S_x be the positive diagonal state-scale matrix listed in table 1. Define the normalized error and

endpoint matrices by

$$\boldsymbol{z} = S_x^{-1}\boldsymbol{e}, \qquad \widehat{A}_i = S_x^{-1}A_iS_x, \quad i \in \{0,1\}. \tag{42}$$

This change of coordinates leaves the endpoint eigenvalues unchanged. It is a numerical choice, not a requirement of the stability proof.

Setting $\boldsymbol{v} = \delta\boldsymbol{\tau}/t_s$ gives the normalized timing-input matrix $B_n = S_x^{-1}\Gamma_\tau t_s$. Discrete LQR on $(\widehat{A}_0, B_n)$ then gives $\boldsymbol{v}^{\mathrm{raw}} = -K_n\boldsymbol{z}$. The corresponding physical-coordinate gain is $K = t_sK_nS_x^{-1}$.

Table 1 lists the weights for two tunings. The aggressive law uses conditioning. The conservative law is applied without conditioning and remains dwell-feasible in this experiment. The table also distinguishes the nominal-design dwell bound from the smaller applied bound $d_{\min}$. The latter is a simulation assumption, not a hardware-calibrated limit.

5.3 Checking the one-cycle model

We first compare the analytical Jacobians with central finite differences of the exact map. Each state step is 10^{-4} times the corresponding diagonal entry of S_x. The timing step is $10^{-3}t_s = 0.01\,\mu\mathrm{s}$. The relative errors are 8.518e-12 for Φ_x and 8.344e-11 for Γ_τ.

Next, we check how the linearization residual changes with perturbation size. We apply simultaneous normalized state and timing perturbations of amplitude ϵ and evaluate

$$\begin{aligned} r_{\mathrm{lin}}(\epsilon) = \big\| S_x^{-1}\big[\mathcal{P}(\bar{\boldsymbol{x}}(0) + \Delta\boldsymbol{x}, \delta\boldsymbol{\tau}) - \bar{\boldsymbol{x}}(0) \\ - \Phi_x\Delta\boldsymbol{x} - \Gamma_\tau\delta\boldsymbol{\tau}\big]\big\|_2. \end{aligned} \tag{43}$$

A second-order remainder should give a slope of two when this residual is plotted against ϵ on logarithmic axes. The fitted slope is 2.000, consistent with that expectation.

5.4 Numerical Lyapunov certificate

For the aggressive gain, we apply (38) to the normalized endpoints $\widehat{A}_0$ and $\widehat{A}_1$. Denote the resulting Lyapunov matrix by P_n and the solver margin by η_n. Julia solves this problem using Convex.jl and Clarabel, with $\mathrm{tr}(P_n) = 3$. The solver returns optimal status and $\eta_n = 2.911\,904\,315 \times 10^{-4}$. The normalized-coordinate matrix, shown to six decimal places, is

$$P_n \approx \begin{bmatrix} 0.608770 & -0.013731 & 0.027354 \\ -0.013731 & 2.384502 & 0.033889 \\ 0.027354 & 0.033889 & 0.006728 \end{bmatrix}. \tag{44}$$

We then check the returned matrix rather than relying on solver status alone. Define the normalized endpoint decrease matrices

$$Q_{i,n} = P_n - \widehat{A}_i^{\mathsf{T}}P_n\widehat{A}_i, \qquad i \in \{0,1\}.$$

Using full precision, their smallest eigenvalues and that of P_n are

$$\begin{aligned} \lambda_{\min}(P_n) &= 4.987\,946\,970 \times 10^{-3}, \\ \lambda_{\min}(Q_{0,n}) &= 2.911\,904\,317 \times 10^{-4}, \\ \lambda_{\min}(Q_{1,n}) &= 2.911\,956\,096 \times 10^{-4}. \end{aligned} \tag{45}$$

Both endpoint decrease margins are positive. Their minimum is $\mu_n = 2.911\,904\,317 \times 10^{-4}$. The reported trace and margins refer to normalized coordinates.

To connect these checks to the physical-coordinate theorem, use

$$\begin{aligned} P &= S_x^{-\mathsf{T}}P_nS_x^{-1}, \\ P - A_i^{\mathsf{T}}PA_i &= S_x^{-\mathsf{T}}Q_{i,n}S_x^{-1}. \end{aligned}$$

These transformations preserve positive definiteness. Thus the numerical checks provide the physical-coordinate certificate required by theorem 1, without changing the feedback gain or the plant trajectories.

The model checks follow a separate sequence. Python reconstructs the physical model and supplies the normalized endpoint matrices to Julia; Julia does not independently reconstruct the converter. The article pipeline compares these endpoints with the normalized matrices from its MATLAB model and aggressive gain. It then recomputes the Lyapunov inequalities using those matrices and the returned P_n.

The machine-readable certificate retains full precision. We also check the rounded matrix displayed in (44) separately. All of these checks use floating-point arithmetic, not interval-arithmetic certification.

5.5 Controlled and open-loop trajectories

All comparison runs start from

$$\boldsymbol{x}_0 = [7.5143 \quad 20.8211 \quad 0.0314]^{\mathsf{T}}, \tag{46}$$

with normalized error 0.983946. Each simulation propagates the plant through its actual dwell durations using exact matrix exponentials. The linearized model is not used in place of the plant. For the continuous-time plots, each interval is sampled at 30 equally spaced local times, including its end. The curves therefore show within-interval motion rather than straight lines between switching states.

Figure 3(a) compares the physical trajectories over 100 cycles. The dot marks the initial state, and the star marks the reference cycle anchor. With feedback, the trajectory rapidly approaches the reference. Open loop remains visibly displaced over this short horizon. Panel (b) extends the comparison and shows that open loop also approaches the same periodic trajectory, but much more slowly.

5.6 Reading the cycle-start error

To compare convergence without mixing it with switching ripple, we sample the state at the start of each cycle. We report the dimensionless error

$$E_k = \left\| S_x^{-1}\big(\boldsymbol{x}(kT) - \bar{\boldsymbol{x}}(0)\big) \right\|_2, \tag{47}$$

where $\bar{\boldsymbol{x}}(0)$ is the cycle anchor. The matrix $S_x = \mathrm{diag}(10, 20, 1)$ uses voltage scales of 10 V and 20 V and a current scale of 1 A. Comparing states at the same phase separates tracking error from the reference's within-cycle ripple. Under the fixed nominal schedule, convergence of these samples to the anchor also implies convergence between samples to the periodic trajectory, not to a constant physical state.

We use $E_k = 0.01$ as a reporting threshold. It is neither a stability boundary nor a hardware accuracy requirement. It is defined relative to the chosen state scales, not to the initial error or the reference components. The condition $E_k < 0.01$ implies component errors below 0.10 V, 0.20 V, and 0.01 A at cycle starts. The joint Euclidean-norm condition is stricter than satisfying these component bounds separately.

Figure 3(b) shows the error histories on logarithmic axes, so cycle zero is omitted. Feedback data extend to cycle 100 and open-loop data to cycle 100000. Errors below 10^{-8} are outside the displayed range. Each marker is an actual sample, not an interpolated crossing. For feedback, it marks the first sample below 0.01. For open loop, it marks the sample after which every remaining simulated error stays below that threshold.

Crossing the threshold does not mean that the error has reached zero. The plot shows its continued decay; the exact open-loop error map provides the separate mathematical guarantee of eventual open-loop convergence.

Table 2 Exact simulation results. Threshold cycles for the two feedback laws denote the first sample with $\|S_x^{-1}\boldsymbol{e}_k\|_2 < 0.01$. The open-loop value denotes the crossing after which every remaining simulated sample stays below 0.01.

Quantity	Value
$\rho(\Phi_x)$, open loop	0.999850
$\rho(A_{\mathrm{cl}})$, aggressive	0.568151
Spectral radius, conservative	0.941468
Minimum β; conditioned cycles	0.293978; 3
Maximum raw offset	65.217 µs
Maximum applied offset	19.172 µs
Minimum raw dwell	-43.045 µs
Minimum applied dwell	3.000 µs
Minimum conservative dwell	15.502 µs
Threshold cycle, conditioned	6
Threshold cycle, conservative	20
Persistent crossing, open loop	22009

5.7 Conditioning during the early transient

Figure 4 places the early error history beside the applied fraction of the raw action and the resulting minimum dwell duration.

The first three requests require factors of approximately 0.294, 0.447, and 0.779. From cycle 3 onward, the complete raw request is feasible and $\beta_k = 1$. Feedback is still active; only the scaling has stopped. The cycle-start error continues to decrease and first falls below 0.01 at cycle 6. Conditioning is transient in this run. This observation does not guarantee that it stays inactive for other initial conditions or after disturbances.

Panel (c) shows why scaling is needed. The aggressive raw law requests a minimum dwell of -43.045 µs, which would invert an interval. These raw actions are candidate commands only. The plant always uses the conditioned schedule.

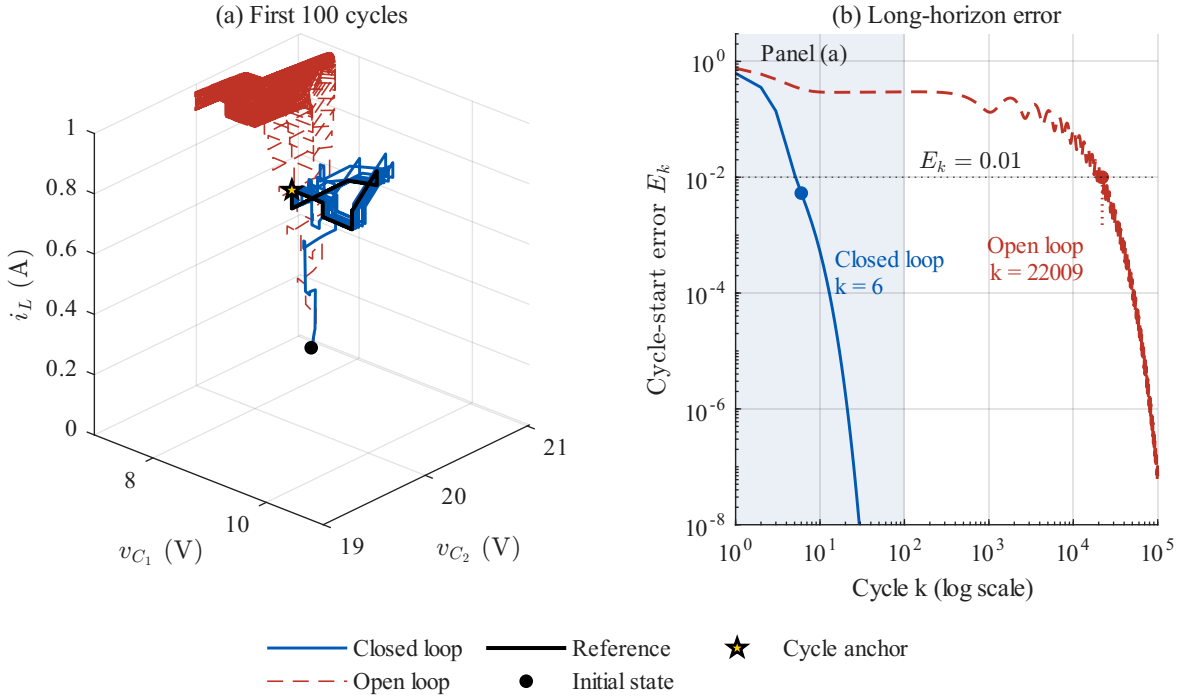


Figure 3 Early transient and eventual convergence. (a) Continuous-time state trajectories over 100 cycles. (b) Normalized cycle-start error on logarithmic axes. Shading marks the horizon in (a); dots mark the saved threshold-crossing samples.

Conditioning acts during 3 cycles, with a minimum factor of 0.293978. It reduces the largest absolute offset from 65.217 to 19.172 µs. Every applied dwell remains at or above 3 µs. When scaling is active, at least one dwell reaches this bound, as the maximal-factor formula predicts.

The conservative law provides a feasible comparison without conditioning. It crosses the error threshold at cycle 20, whereas the conditioned aggressive law crosses at cycle 6. Table 2 collects these values. The exact nonlinear simulations demonstrate performance for this benchmark and initial condition. The common-P theorem alone does not certify their entire large-error transient.

5.8 Physical states and long-horizon comparison

Figure 5 shows each voltage and current against a common time axis. Both responses start from the same state. The reference curve repeats the nominal within-cycle waveform rather than its average or a constant anchor value. The closed-loop states approach this waveform while open loop remains displaced over the displayed window.

The continuing switching ripple belongs to the reference motion. It is not residual cycle-start tracking error. These curves retain the exact within-interval samples, so they show motion that cycle-start samples alone would omit.

The short window hides the eventual open-loop convergence. With fixed nominal durations, (26) gives the exact affine map $\mathcal{P}(\boldsymbol{x}, \mathbf{0}) = \Phi_x \boldsymbol{x} + \boldsymbol{h}$. Subtracting the fixed anchor $\bar{\boldsymbol{x}}(0)$ cancels the constant term, leaving

$$\boldsymbol{e}_{k+1} = \Phi_x \boldsymbol{e}_k \tag{48}$$

without a linearization remainder. For this schedule, $\rho(\Phi_x) = 0.999850 < 1$, so open loop converges throughout the ideal affine model. This conclusion relies on fixed durations and does not certify the nonlinear timing-feedback map.

The separate 100000-cycle simulation in figure 3(b) checks this longer time scale. Its error stays below 0.01 from cycle 22009 through the end of the run. With $T = 286\,\mu s$, the open-loop

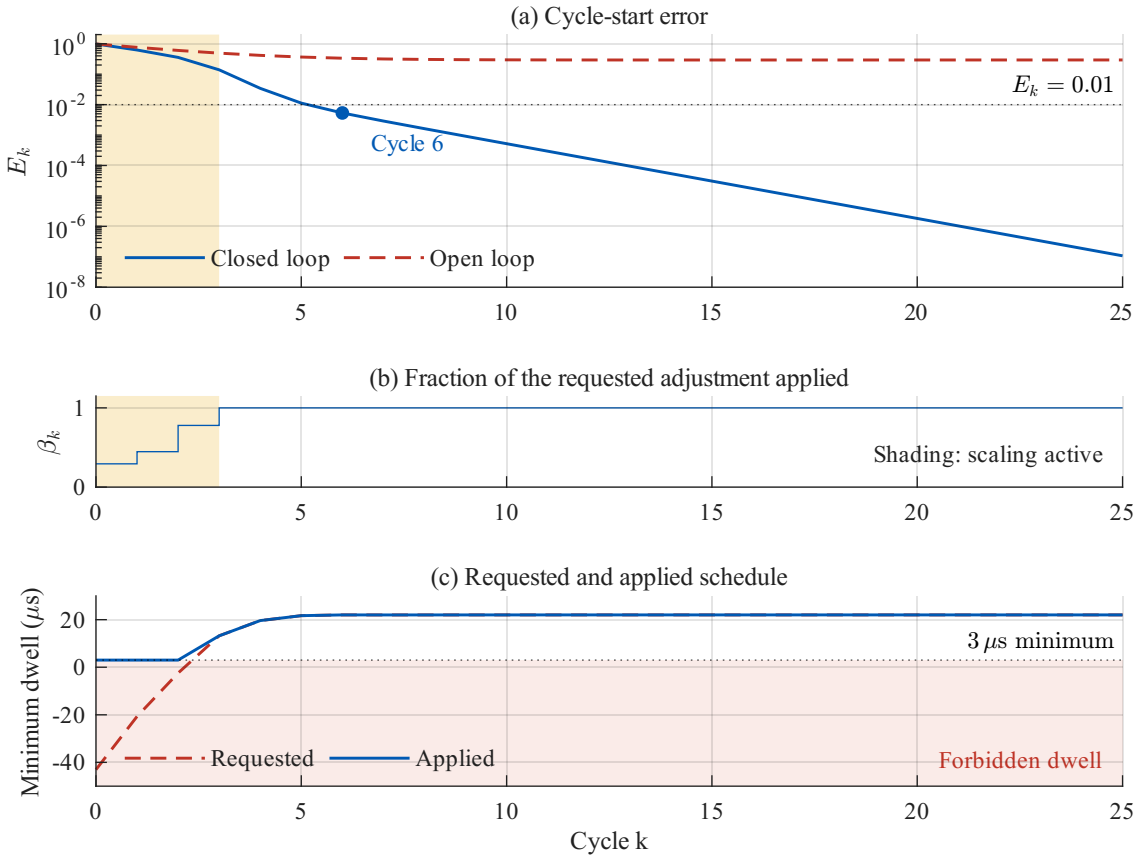


Figure 4 First 25 cycles of the exact simulation. (a) Cycle-start error. (b) Applied fraction of the timing request. Shading in (a)–(b) marks active scaling. (c) Requested and applied minimum dwell, with the forbidden range shaded.

transient spans seconds. The controlled threshold crossing occurs within a few milliseconds.

5.9 Host computational cost

We measured the 95th-percentile execution time, denoted p95, for computing the conditioning factor β. After warm-up, we timed 10 000 calls using the saved simulation actions as inputs.

The measurements ran on a desktop computer with an AMD Ryzen 7 7800X3D 8-Core Processor, using MATLAB R2025b and a double-precision C++ implementation compiled with GCC 13.2.0 and `-O3` optimization.

In MATLAB, computing β alone took 3.6 µs at p95. Including offset scaling, dwell reconstruction, and feasibility checks increased the full-conditioner p95 to 5.3 µs.

In C++, the individual-call p95 reached the timer baseline of 0.1 µs, so individual execution times could not be resolved reliably. We therefore also timed batches of 200 calls. The p95 of the batch-average cost was 7.5 ns per call for β alone and 25.5 ns for full conditioning. These batch statistics are not individual-call latency percentiles.

All reported timings were measured on a desktop-class computer, not an embedded system. They exclude acquisition and waveform generation and do not establish execution times or deadline guarantees for embedded hardware.

6 Discussion

The results separate dwell feasibility from convergence. For any finite raw action, the dwell inequalities give the largest feasible scale directly.

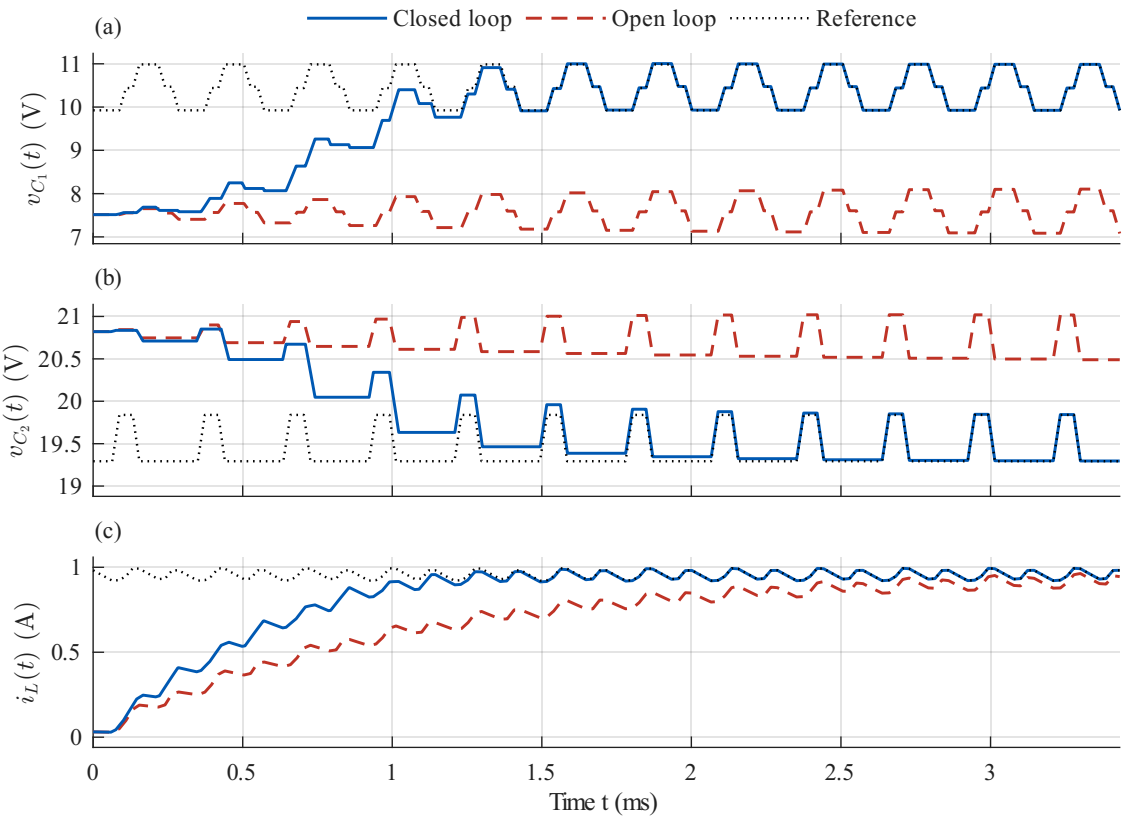


Figure 5 Continuous-time state responses over the first 12 cycles. (a) $v_{C_1}(t)$. (b) $v_{C_2}(t)$. (c) $i_L(t)$. Each panel compares closed loop, open loop, and the reference periodic waveform.

The Lyapunov inequalities test whether the associated linearized dynamics converge as that scale changes. Their common-P certificate is sufficient for uniform exponential stability, but it is not a global nonlinear guarantee.

Neither calculation adds an online optimization problem. The SDP and the supporting invariant-region checks run offline. At each cycle start, the controller only multiplies the error by a fixed gain, computes adjacent differences, and selects the smallest upper bound on the factor.

The method builds on established results. The one-cycle sensitivity follows Patino et al. Patino et al. (2010). Its common-state-matrix reduction recovers Marcolino et al.'s fixed-period timing model Marcolino et al. (2021). Direction-preserving scaling also appears in control allocation Nakamura-Zimmerer and Miller (2026). Here, the contribution is to obtain that scale in closed form for coupled dwell inequalities and connect the resulting timing feedback to a common-P certificate. We do not propose a new LQR or a new general Lyapunov theorem.

The certificate has two main restrictions. It uses one quadratic function over the whole interval, and it includes the zero-feedback endpoint. It can therefore fail even when a more restricted controller is stable. In this benchmark, open loop already converges to the reference. Feedback improves convergence speed while preserving feasible schedules.

The exact nonlinear simulations demonstrate performance for the chosen initial condition. They do not certify a nonlinear region of attraction.

Hardware effects are also outside the study. The assumed 3 µs bound has not been calibrated experimentally. We do not model timer quantization, delays, dead time, or implementation margin.

State estimation, parameter uncertainty, and experimental validation remain to be addressed. The mode sequence, period, and reference phase are fixed throughout.

7 Conclusion

The proposed conditioner finds the largest dwell-feasible scalar multiple of a proportional timing action with an $O(N)$ scan. It preserves the action's direction, mode order, and cycle period without an online solver.

An offline endpoint SDP supplies a common quadratic Lyapunov function. This function certifies uniform exponential stability of the conditioned linearized model for every factor sequence in $[0, 1]$.

In the converter simulations, conditioning makes the aggressive requests feasible and gives faster convergence than the conservative law and the already stable open-loop schedule. The long-horizon comparison shows the much slower open-loop approach to the same periodic motion. A nonlinear convergence certificate for the full controlled transient requires further analysis.

Appendix A Feasible-region expansion under timing-action scaling

A.1 Expansion of the dwell-feasible region

A smaller fixed factor allows larger errors while keeping the timing action feasible. To see this, hold $\beta \in (0, 1]$ fixed and apply $\delta\boldsymbol{\tau} = -\beta K\boldsymbol{e}$. Substitution into the dwell inequalities gives the feasible region

$$\begin{aligned} \mathcal{F}_\beta &= \{\boldsymbol{e} : \beta DK\boldsymbol{e} \le \boldsymbol{m}\} = \frac{1}{\beta}\mathcal{C}, \\ \mathcal{F}_1 &= \mathcal{C}. \end{aligned} \tag{A1}$$

At $\beta = 1$, this is the Raw-Action Admissible Region $\mathcal{C}$. Reducing β expands the region about the origin by a factor of $1/\beta$ in every direction. Thus $0 < \beta_a < \beta_b \le 1$ implies $\mathcal{F}_{\beta_b} \subseteq \mathcal{F}_{\beta_a}$.

For the three-state benchmark, the volume grows by β^{-3}. The comparison factors 0.25, 0.5, and 1 give volume ratios 64, 8, and 1, respectively. This geometric expansion follows directly from the dwell inequalities. It describes feasibility in the current cycle, not stability over successive cycles.

Figure A1 compares the nested regions with errors from the conditioned simulation. The plotted trajectory uses the maximal state-dependent factor at each cycle. It does not hold any one comparison factor fixed.

The initial error lies inside $\mathcal{F}_{0.25}$ but outside $\mathcal{F}_{0.5}$. This agrees with the first applied factor of approximately 0.294. The raw action becomes feasible at cycle 3.

Lines in the figure join cycle-start samples, not within-cycle trajectories. The planar panels are projections, not slices with one error component set to zero. We determine membership and entry from the full three-dimensional inequalities, not from the projections.

A.2 Feasibility over successive cycles

An error may produce a feasible action now without remaining in the same feasible region at the next cycle. To check feasibility over successive cycles, we keep each comparison factor β fixed. The corresponding linearized physical-error dynamics are

$$\boldsymbol{e}_{k+1} = M_\beta \boldsymbol{e}_k, \qquad M_\beta = \Phi_x - \beta\Gamma_\tau K. \tag{A2}$$

The region is positively invariant if this map sends every point in it back into the region. For each of the three plotted factors, we checked

$$M_\beta \mathcal{F}_\beta \subseteq \mathcal{F}_\beta. \tag{A3}$$

The calculation runs offline in MPT3 Herceg et al. (2013), using the preimage-containment approach of Gilbert and Tan (1991). All three regions pass for this benchmark. Equivalently, every vertex $\boldsymbol{v}$ satisfies $\beta DKM_\beta \boldsymbol{v} \le \boldsymbol{m}$ within numerical tolerance. The benchmark regions are bounded polyhedra, so linearity and convexity extend the vertex checks to all their points. Repeated application of the same fixed-factor controller therefore keeps its linearized trajectory dwell-feasible.

Invariance gives continued feasibility. Convergence follows separately from theorem 1, because $M_\beta = A(\beta)$ in physical error coordinates. Together,

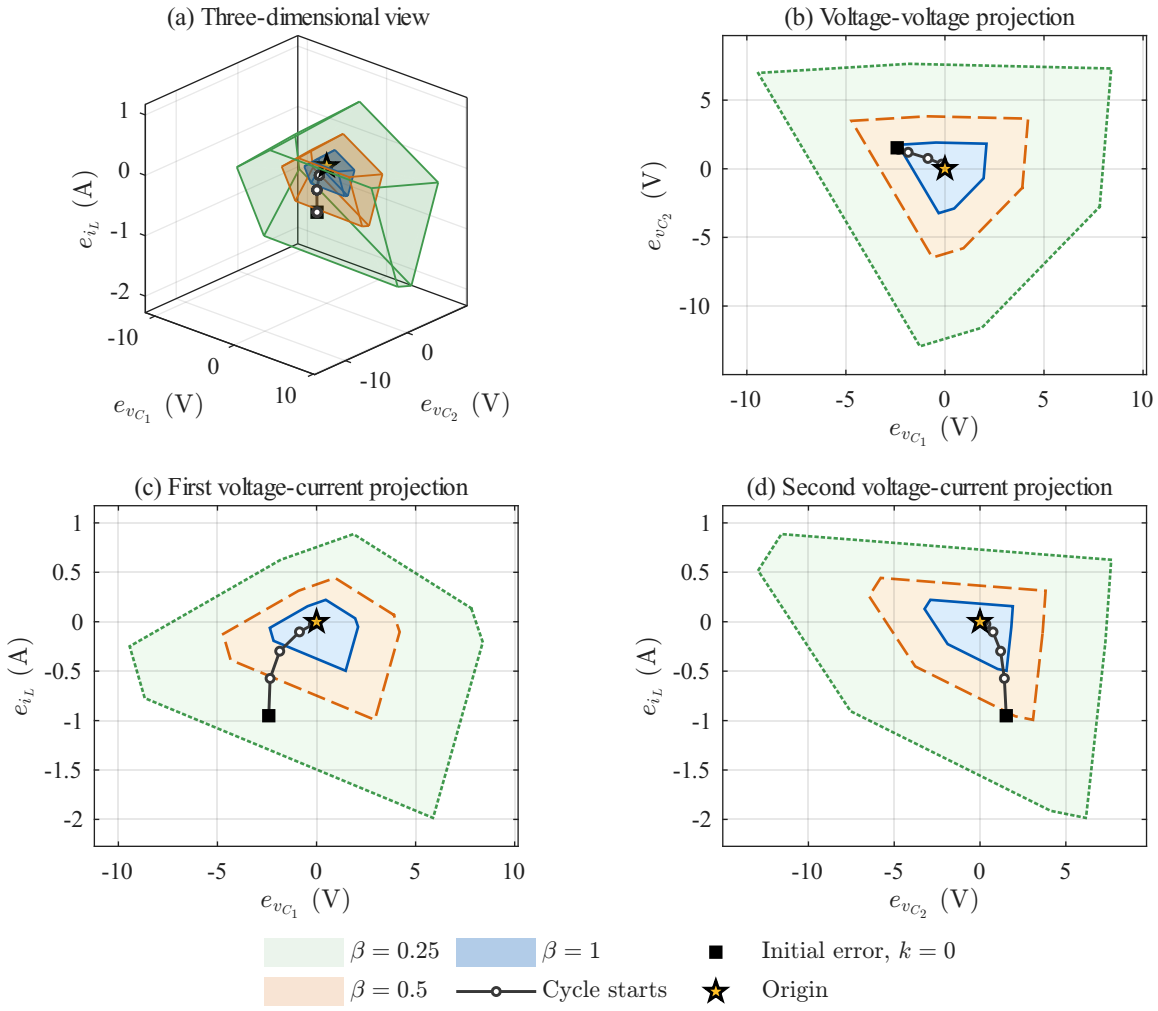


Figure A1 Dwell-feasible regions for fixed factors $\beta = 0.25, 0.5, 1$, with early cycle-start errors from the conditioned simulation. (a) Three-dimensional view. (b)–(d) Coordinate-plane projections. Smaller factors admit larger errors.

these results show that each plotted region admits feasible, convergent linearized operation under its corresponding fixed factor. At $\beta = 1$, this is the raw controller on $\mathcal{C}$. Smaller factors admit larger regions for their respective controllers.

These checks do not prove the same invariance for the exact nonlinear plant. They also use fixed factors, whereas the simulated controller chooses β_k from the current error. The plotted expansion is therefore not an enlarged nonlinear region of attraction. The displayed large-error trajectory remains simulation evidence. The common-P theorem covers the conditioned linearized family without relying on these polyhedral checks.

Declarations

Funding. TODO: name the funding sources and grant numbers, or state that no funding was received.

Competing interests. TODO: state whether the authors have relevant financial or non-financial competing interests.

Data availability. TODO: state where the generated simulation data are available, or that all

data supporting the findings are contained in the article. Required for original research.

Code availability. TODO: state where the simulation and figure-generation scripts are available, or that this item is not applicable.

Author contributions. TODO: describe the contribution of each author.

Acknowledgements. TODO: add if applicable, otherwise delete this heading.

ORCID. TODO: add the author ORCID identifiers if available.

References

Augustine, M.T., Patil, D.U.: A practically stabilizing model predictive control scheme for switched affine systems. IEEE Control Systems Letters **7**, 625–630 (2023) https://doi.org/10.1109/LCSYS.2022.3210368

Albea-Sanchez, C., Sferlazza, A., Gómez-Estern, F., Gordillo, F.: Control of power converters with hybrid affine models and pulse-width modulated inputs. IEEE Transactions on Circuits and Systems I: Regular Papers **68**(8), 3485–3494 (2021) https://doi.org/10.1109/TCSI.2021.3083900

Oliveira, M.C., Geromel, J.C., Hsu, L.: LMI characterization of structural and robust stability: The discrete-time case. Linear Algebra and its Applications **296**(1–3), 27–38 (1999) https://doi.org/10.1016/S0024-3795(99)00086-5

Egidio, L.N., Daiha, H.R., Deaecto, G.S.: Global asymptotic stability of limit cycle and H_2/H_∞ performance of discrete-time switched affine systems. Automatica **116**, 108927 (2020) https://doi.org/10.1016/j.automatica.2020.108927

Egidio, L.N., Deaecto, G.S., Jungers, R.M.: Stabilization of rank-deficient continuous-time switched affine systems. Automatica **143**, 110426 (2022) https://doi.org/10.1016/j.automatica.2022.110426

Flieller, D., Riedinger, P., Louis, J.-P.: Computation and stability of limit cycles in hybrid systems. Nonlinear Analysis: Theory, Methods & Applications **64**(2), 352–367 (2006) https://doi.org/10.1016/j.na.2005.06.054

Gilbert, E.G., Tan, K.T.: Linear systems with state and control constraints: The theory and application of maximal output admissible sets. IEEE Transactions on Automatic Control **36**(9), 1008–1020 (1991) https://doi.org/10.1109/9.83532

Herceg, M., Kvasnica, M., Jones, C.N., Morari, M.: Multi-parametric toolbox 3.0. In: Proceedings of the European Control Conference, Zürich, Switzerland, pp. 502–510 (2013). http://control.ee.ethz.ch/~mpt

Hu, T., Lin, Z., Chen, B.M.: Analysis and design for discrete-time linear systems subject to actuator saturation. Systems & Control Letters **45**(2), 97–112 (2002) https://doi.org/10.1016/S0167-6911(01)00168-2

Karamanakos, P., Liegmann, E., Geyer, T., Kennel, R.: Model predictive control of power electronic systems: Methods, results, and challenges. IEEE Open Journal of Industry Applications **1**, 95–114 (2020) https://doi.org/10.1109/OJIA.2020.3020184

Marcolino, M.H., Galvão, R.K.H., Kienitz, K.H.: Predictive control of linear systems with switched actuators subject to dwell-time constraints. Journal of Control, Automation and Electrical Systems **32**(1), 1–17 (2021) https://doi.org/10.1007/s40313-020-00667-9

Mason, O., Shorten, R.: On common quadratic Lyapunov functions for stable discrete-time LTI systems. IMA Journal of Applied Mathematics **69**(3), 271–283 (2004) https://doi.org/10.1093/imamat/69.3.271

Nakamura-Zimmerer, T., Miller, L.J.: Robust direction-preserving control allocation. In: AIAA SciTech 2026 Forum (2026). https://doi.org/10.2514/6.2026-2353

Olalla, C., Leyva, R., El Aroudi, A., Queinnec, I.: Robust LQR control for PWM converters: An LMI approach. IEEE Transactions on Industrial Electronics **56**(7), 2548–2558 (2009) https://doi.org/10.1109/TIE.2009.2017556

Patino, D., Riedinger, P., Ruiz, F.: A predictive control approach for DC–DC power converters and cyclic switched systems. In: 2010 IEEE

International Conference on Industrial Technology, pp. 1259–1264. IEEE, ??? (2010). https://doi.org/10.1109/ICIT.2010.5472624

Saeed, J., Wang, L., Fernando, N.: Model predictive control of phase shift full-bridge DC–DC converter using Laguerre functions. IEEE Transactions on Control Systems Technology **30**(2), 819–826 (2022) https://doi.org/10.1109/TCST.2021.3069148

Xu, D., Lazar, M.: Finite Control Set Model Predictive Control with Limit Cycle Stability Guarantees (2024). https://doi.org/10.48550/arXiv.2407.07615

Yang, Q., Karamanakos, P., Tian, W., Gao, X., Li, X., Geyer, T., Kennel, R.: Computationally efficient fixed switching frequency direct model predictive control. IEEE Transactions on Power Electronics **37**(3), 2761–2777 (2022) https://doi.org/10.1109/TPEL.2021.3114979